Olist E-Commerce

Data Analysis Report

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Executive Summary

This report presents an end-to-end exploratory data analysis of over 100,000 real Brazilian e-commerce orders sourced from the Olist platform (2016–2018). The analysis covers sales trends, category-level profitability, customer segmentation via RFM methodology, delivery performance, and geospatial distribution of demand across Brazil.

Three critical business insights emerged from the analysis:

- Order volume and profitability are heavily concentrated in the São Paulo–Rio de Janeiro corridor, revealing untapped potential in northern and inland regions.
- The majority of customers have not placed an order in over 300 days, indicating a retention problem. Champions and Loyal segments, though small, drive disproportionate revenue.
- Delivery times peak around 7–10 days (median: 10 days) but show a heavy right tail beyond 100 days. Late deliveries — likely concentrated in remote regions — are a key driver of low review scores and customer churn.

Dataset Overview

The analysis is based on the Olist Brazilian E-Commerce Public Dataset, available on Kaggle. It consists of 8 relational tables covering orders, customers, products, sellers, payments, reviews, and geolocation data.

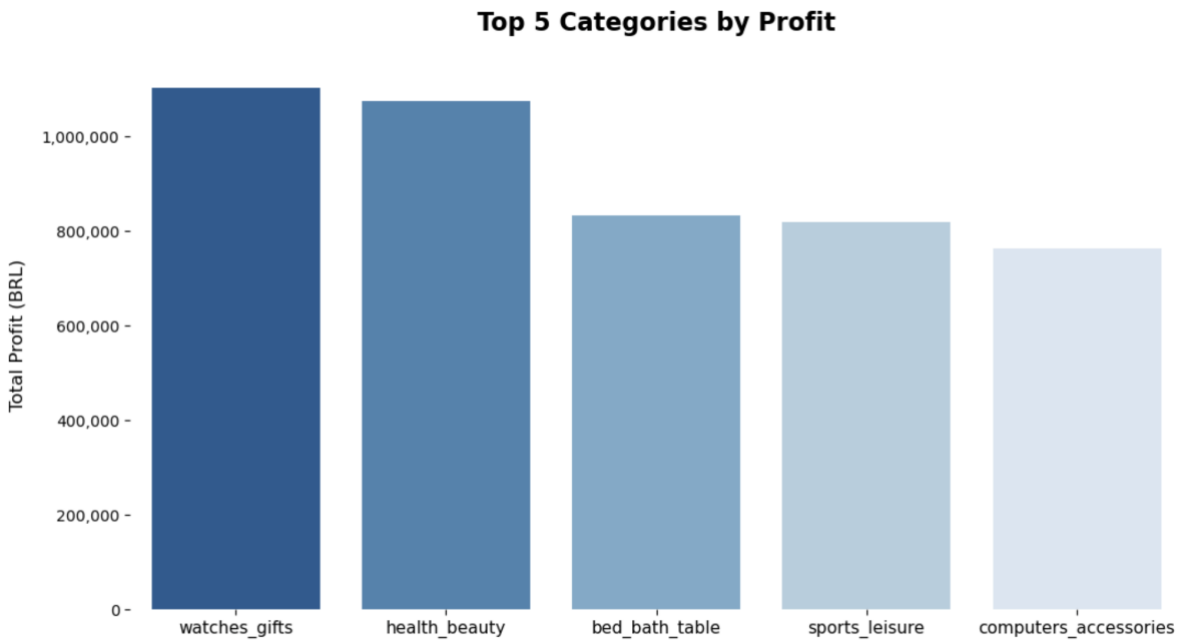
Table	Rows (approx.)	Description
orders	~100K	Order status & timestamps
customers	~100K	Customer location data
payments	~100K	Payment method & value
order_items	~112K	Product & freight details
products	~32K	Product categories
reviews	~99K	Review scores & comments
sellers	~3K	Seller location data
geolocation	~1M	ZIP code coordinates

1. Monthly Sales Trend

Insight

Sales show a strong upward trajectory from mid-2017, peaking in early 2018. The growth pattern suggests effective seasonal campaigns and growing platform adoption. A sharp drop at the tail of the data is consistent with the dataset cutoff rather than a real decline.

2. Category-Level Profitability

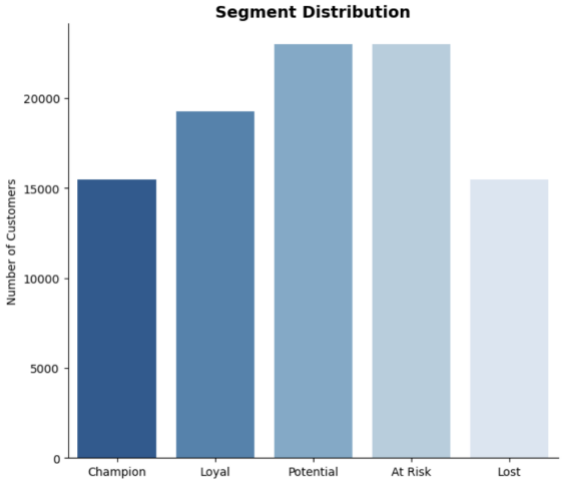


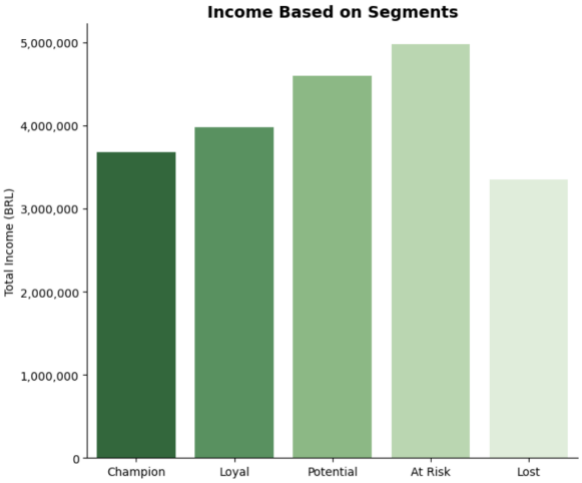
Insight

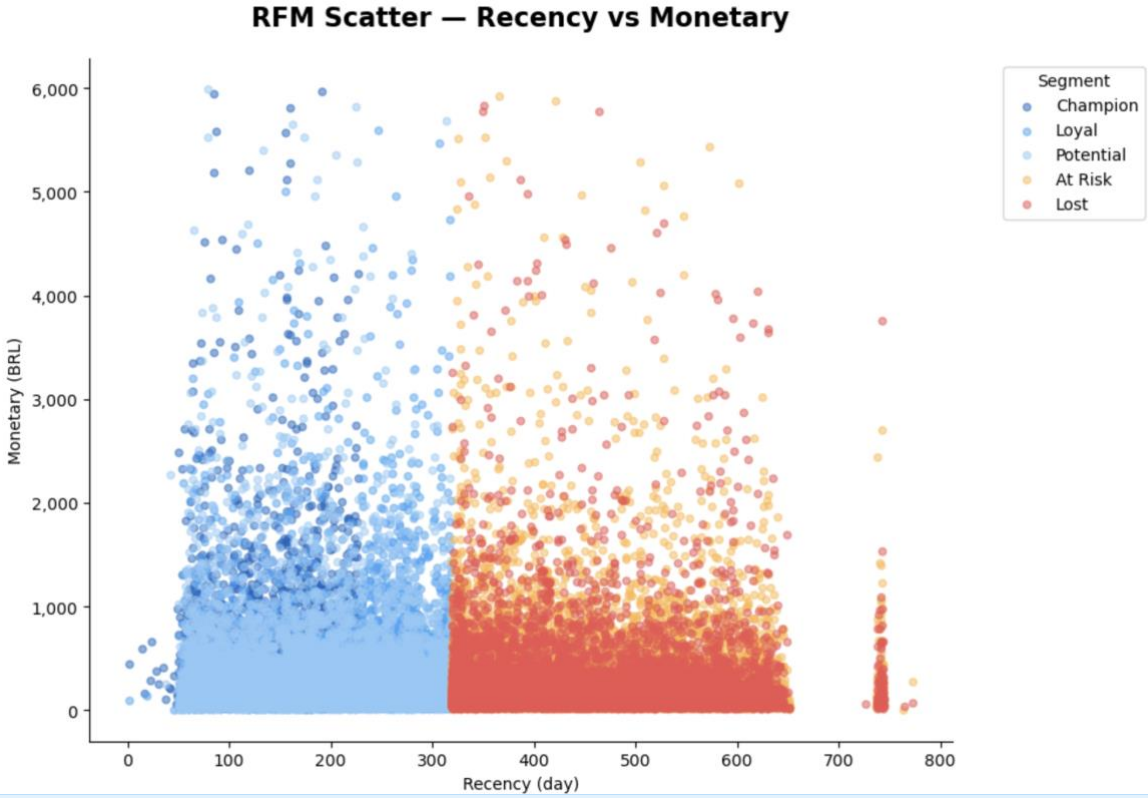
A small number of product categories drive the majority of platform profit. The top 5 categories account for a disproportionate share of total revenue, suggesting that inventory and marketing efforts should be concentrated in these high-margin segments to maximize ROI.

3. RFM Customer Segmentation

Customers were segmented using the RFM (Recency, Frequency, Monetary) framework. Each customer was scored on a 1–5 scale across all three dimensions and assigned to one of five segments.



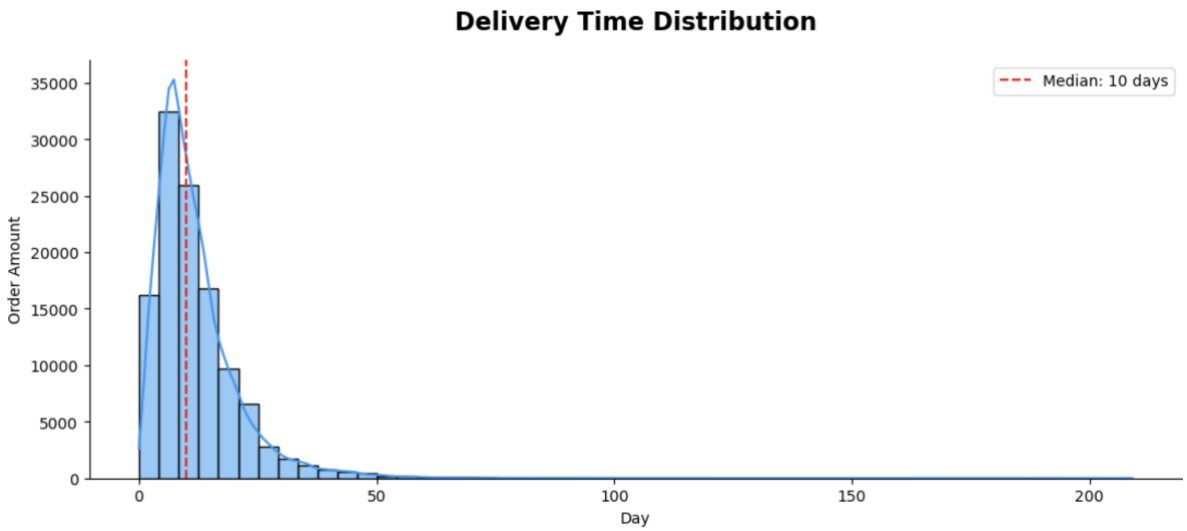




Insight

The majority of customers fall into the At Risk and Lost segments, having not purchased in 300+ days. Champions and Loyal customers, concentrated in the lower-left of the scatter plot (recent & high-spend), represent the highest lifetime value. Targeted retention campaigns for At Risk customers could recover significant revenue at low acquisition cost.

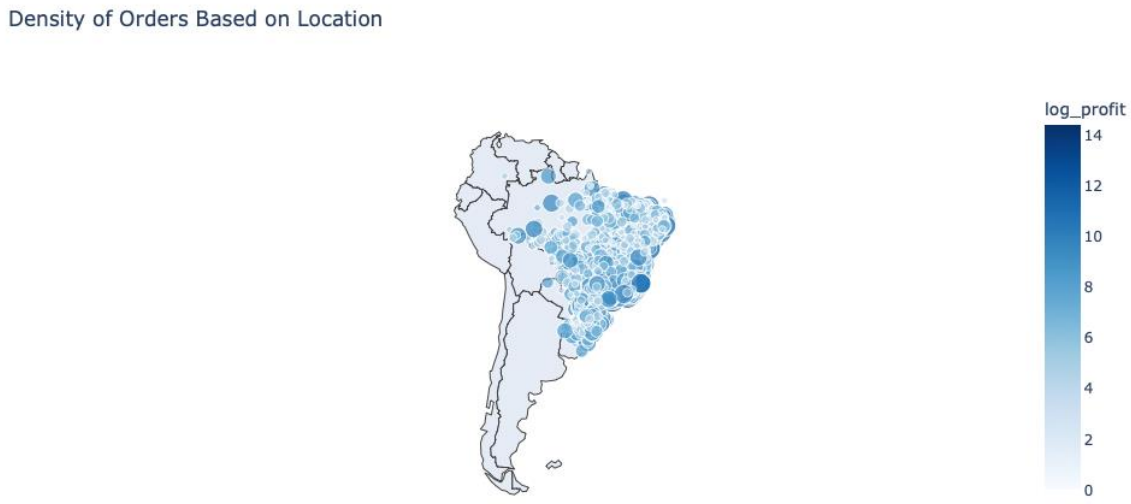
4. Delivery Time Distribution



Insight

Delivery times peak sharply around 7–10 days, with the median at 10 days. However, the distribution has a heavy right tail extending well beyond 100 days, indicating a meaningful subset of severely delayed orders. The sharp peak followed by a long tail suggests that while most deliveries are handled efficiently, a significant minority face extreme delays — likely concentrated in remote northern and inland regions where logistics infrastructure is weaker. These late deliveries are a primary driver of low review scores and customer churn.

5. Geospatial Order Density



Insight

Order volume and profitability are heavily concentrated along the southeastern coastal corridor, particularly around São Paulo and Rio de Janeiro. Northern and inland regions show significantly lower penetration, representing an untapped growth opportunity that could be unlocked through targeted logistics partnerships and regional marketing strategies.

Conclusions & Recommendations

1. Invest in Customer Retention

The RFM analysis reveals a critical retention gap. A targeted win-back campaign for At Risk customers — using personalized discounts or reminders — is the highest-ROI short-term action available.

2. Focus on High-Margin Categories

Category profitability is highly concentrated. Marketing spend and seller acquisition efforts should prioritize the top 5 categories identified in this analysis to maximize platform-level returns.

3. Optimize Logistics for Distant Regions

The delivery time distribution and geospatial map together suggest that customers in northern and inland Brazil experience longer, more variable delivery times. Partnering with regional logistics providers could simultaneously reduce delivery times, improve review scores, and open new market segments.

Tools & Technologies

- Python (Pandas, NumPy) — data cleaning and transformation
- pandasql — SQL queries on DataFrames (window functions, GROUP BY)
- Seaborn & Matplotlib — static visualizations
- Plotly — interactive geospatial map
- Jupyter Notebook — development environment

Dataset: Olist Brazilian E-Commerce Public Dataset — kaggle.com/datasets/olistbr/brazilian-ecommerce